Fiscal Year:	FY 2010	**Task Last Updated:**	FY 04/13/2010
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chaikin@NYU.edu	**Fax:**	FY 212 995-4016
PI Organization Type:	UNIVERSITY	**Phone:**	212 998-7694
Organization Name:	New York University		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10003	**Congressional District:**	8
Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
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No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Motil, Brian J.	**Contact Phone:**	216-433-6617
Contact Email:	brian.j.motil@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hollingsworth, Andrew (New York University)		
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	Colloid science is entering a new era. Over the past 15 years, our NASA-sponsored research has mainly dealt with monodisperse suspensions of colloidal particles interacting via well-known forces. Using spherical particles and observations with light scattering and microscopy, we have gained a great deal of fundamental knowledge about different phases of matter and the dynamics and thermodynamics of their formation. In particular, our experimental results in microgravity have lead to a basic understanding of why crystals and glasses form and their properties. During the past decade, we have made great strides in synthesizing new classes of particles with different shapes and specific, reversible or irreversible, variable range interactions. We have also found new ways to manipulate the particles with flow, electric and magnetic fields and light. We are therefore positioned at the threshold of a new technology, assembling equilibrium and non-equilibrium macroscopic structures with function and activity from well designed		

Task Description:	particles on the nano to micron scale. Of course, there are still fundamental scientific questions which we can and will address including a host of new ordered phases, frozen configurations, frustration and glasses and the process of self-organization itself. In particular, we plan to use the microscopy and light scattering instruments, in collaboration with our European colleagues, to study particles that we prepare through emulsion and dispersion polymerization. Physical lithographic techniques will also be employed, and the particles will be modified chemically for controllable interactions. We plan to use different phoretic techniques– electro-, dielectro-, and thermo-phoresis– to control the particles density and orientation. These will also serve as the driving forces to establish the rheological properties of these new systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Characterization of crystal formation in the microgravity environment of the ISS (International Space Station) can lead to a greater understanding of how gravity affects many kinds of colloidal materials, including monodisperse ellipsoids and cubes, colloidal clusters of silica or polymer microspheres, DNA-functionalized colloidal spheres, and 'lock-and-key' colloids. By performing these experiments in reduced gravity, we intend to accomplish the desired characterization without gravitationally-induced inhomogeneities that affect both the dynamics and equilibrium state on earth. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices to be used in optical communication systems. With the ability to make particles of different shapes, i.e., non spherical, we also have the possibility of having directionally dependent particle interactions. For example, we could take tetrahedral clusters of particles and attach DNA to them. The complementary single-stranded DNA 'sticky ends' can associate/dissociate via thermal activation. This arrangement could lead to tetrahedral bonding as found in diamond or in amorphous glass structures. Another approach utilizes depletion interactions. Since we can lithographically prepare particles of any shape we design in two dimensions and many shapes in three dimensions, we can fabricate lock-and-key colloids which only bind to their complementary shape. In this case, the binding is also directional since the congruent surfaces must match. We can also make such lock-and-key particles through emulsion chemistry. Our goal is to produce some simple processes with such ‘designer particles’ and interactions, to lay the foundations for self-assembly and perhaps self-replication of this new class of materials.
Task Progress:	Flight projects BCAT-5, BCAT-6 BCAT-5 (Binary Colloidal Alloy Test – 5) is a follow-on experiment to BCAT-3, which has been performed during Increments 8, 9, 10, 12 and 13 and is scheduled to run on 16 and 17, and BCAT-4 (launched March 2008). BCAT-5 was launched on July 15, 2009 aboard the Space Shuttle Endeavour and includes ten colloidal samples. NYU prepared one of these (Sample 9) consisting of poly(methyl methacrylate) (PMMA) particles whose size and concentration are tuned to increase the rate at which crystals nucleate using spherical seed particles. The results of this experiment have important consequences for the control of the resulting crystal size distribution. BCAT-5 represents a significant opportunity to explore the fundamental physics behind these processes by crystallizing in microgravity where there is no gravitational jamming to suppress the natural ordering process observed in microgravity. NYU also prepared PMMA particles for PIs Frisken and Bailey of Simon Fraser University (BCAT-5 Samples 6, 7, 8). These particles are used to study the competition between phase separation and crystallization. BCAT-6 samples 9 and 10, seeded growth, will build-on our NASA PHaSE experiment that showed that glasses in concentrated systems (as seen on Earth) tend to order into crystals when gravitational jamming is removed. BCAT-6 is scheduled to launch on one of the final space shuttle missions. Monodisperse particles have been created for this sample, which are the size of the wavelength of light. White light from a flashlight and the camera flash will be used to detect the colorful Bragg reflections produced by colloidal crystals formed following the sample mixing. What’s new and significant about this sample is the idea of applying spherical seed particles to the wall of the sample cell. In this experiment, half of the surface is covered by seeds, while the other portion of the container is not. The PMMA colloidal spheres were synthesized by dispersion polymerization using a poly(12-hydroxystearic acid)-g-poly(methyl methacrylate) stabilizer prepared in this lab. BCAT-6 sample 12, PS-DNA, represents the first micro-gravity flight of self-annealing, DNA-coated polystyrene colloids. Thermally induced hybridization of the constituent linker strands will allow the particles to bind together, thus enabling them to explore self-assembly, reproduction, fidelity, and mutation. Changes in cabin temperature of few degrees should cause the aggregated particle structures to melt (unbind) as the ISS moves in the view of the sun; the individual particles will subsequently recombine as the view becomes blocked by the earth and the ambient temperature decreases. We will look for changes in sample turbidity and Bragg reflections produced by any crystallites that may form during the thermal cycling. Samples will be returned to earth for microscopic observation. And in future experiments, we will leverage this work to exploit how they offer exponential growth-needed for next stage of micro- and nano-technology with one trillion component pieces. In addition to providing an expanded toolkit to colloidal engineering, exponential growth provides information on competition and evolution, which can then be used to enhance colloidal self-assembly models. Our BCAT 6 sample 11, silica ellipsoids in a refractive index matching solvent, was disqualified due to solvent compatibility issues with the sample module materials. Samples for PACE-2 In preparation for the Preliminary Advanced Colloids Experiments (PACE), we synthesized two types of aspherical colloids. This class of colloidal particles includes silica ellipsoids and cubes, SEM images of which were taken at Princeton for characterization analysis. We have found that depending on the particular shape, ellipsoidal particles can pack more densely than spheres in both the crystalline and random configurations. What is particularly interesting is that the difference in packing fraction between random and crystal is smaller for ellipsoids than for spheres; thus, the driving force for crystallization is reduced. In fact, following our work on granular ellipsoids, several groups have numerically tried to find a fluid-crystal transition from computer simulations– as of yet, there have been no reports of crystallization. While simulations and experiments with granular materials can provide information about packing they do not let us investigate the thermodynamic system the same way that experiments on colloidal systems will. Regarding the colloidal cubes, we have done experiments that show that crystallization can be induced with depletion agents (dissolved polymers); however, the micron-size silica

	cubes are not density matched with the suspending medium. We will eliminate the dissolved polymer and study the crystallization and phase behavior of a concentrated sample in micro-gravity. Preliminary tests will be conducted using a rapid prototyped cell filled with monodisperse hollow silica (core-shell) ellipsoids (three concentrations) and silica cubes (two types). Microscope specialty cells (PACE-2 meeting January 15, 2010 at NYU) Preliminary Advanced Colloids Experiment (PACE) Advanced Colloids Experiment (ACE) Meeting attendees: Bill Meyer (NASA GRC); Ron Sicker (NASA GRC project manager); Tony Bruzas (Zin Technologies task lead engineer); John Eustice (Senior Scientist, Zin Technologies); Paul Chaikin (NYU); Andrew Hollingsworth (NYU) During the one-day meeting at NYU, we discussed test sample requirements, preparation, and delivery dates. We also presented an update, with laboratory demonstrations, on specialty cell development. PACE-2 will be used to characterize the capability of conducting high magnification imaging with LMM using a test target and test particles supplied by NYU and other research groups. This will be done to find the minimum size particles are needed for ACE, given microscope vibration and magnification results. Specialty cell development A temperature gradient cell for microscopy has been developed to grow large (size ~ 3 mm) colloidal single crystals [see Cheng et al., Nature 401 (1999) 893–895] and also to study the reversible aggregation behavior of DNA-linked colloids [see Valignat et al., Proc. Nat. Acad. Sci. 102 (2005) 4225–4229]. Thermal data was collected for different source potentials. Typical temperature gradients in the sample range from 0.2 to 5 deg. C/mm, depending on the source voltage and sample cell configuration. The E-field cell, also known as the “electric bottle”, was developed in the Chaikin-Russel lab for microgravity research. Further refinements to the cell design will allow us to change and control in situ particle concentration. The measurement of spatially dependent particle concentration, together with a known electric field profile, will allow the osmotic pressure of the suspension to be determined as a function of particle volume fraction. We plan to use confocal microscopy to measure the concentration profiles by tracking and counting colloidal particles across the sample.
Bibliography Type:	Description: (Last Updated: 09/29/2025)
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Articles in Peer-reviewed Journals	Sacanna S, Irvine WTM, Chaikin PM, Pine DJ. "Lock and key colloids." Nature. 2010 Mar 25;464(7288):575-8. PMID: 20336142 , Mar-2010
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Papers from Meeting Proceedings	Sacanna S, Irvine WTM, Chaikin PM, Pine DJ. "Lock-and-key colloids through polymerization-induced buckling of monodispersed silicon oil droplets." Meeting of the American Physical Society, Portland, OR, March 15–19, 2010. Bulletin of the American Physical Society, Vol. 55, 2010. , Mar-2010
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